Gravity as Relational Difference Elimination: A Harmonic Closure Derivation

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Model status. This manuscript is an exploratory, falsifiable framework. Claims are labeled throughout using the taxonomy in Table 1. No experimental validation independent of the author's prior program is claimed. A relativistic completion does not yet exist. Version 2.3 establishes that the framework defines a natural gravitational scale $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$ constructed entirely from hadronic constants and the $n = 4$ closure, and identifies α as the relational ratio of the visible mode within that closure: $\alpha = \alpha_G\,\omega_V^2/(\omega_N\omega_m)$. G and α are not independently free — they are linked through this scale.

Claim Type	Description
[POSTULATE]	A foundational assumption — not derived, not yet falsified.
[DERIVED]	A mathematical consequence of the postulates.
[CALIBRATION]	A parameter fixed by matching to an empirical value.
[HYPOTHESIS]	A testable physical conjecture requiring external validation.
[SPECULATIVE]	An exploratory program note. Not yet a genuine prediction.

Table 1: Claim-type legend used throughout the manuscript.

Abstract

We propose a relational–informational model in which gravity emerges as the macroscopic tendency of physical systems to eliminate differences between projected relational states. The framework is based on a universal harmonic partition of system frequencies $\omega_N^2 = \omega_V^2 + \omega_m^2$, previously applied within the same research program to hadronic, atomic, and cosmological systems.

The distance between two systems is identified as a *relational phase offset* — the accumulated difference in identity-mode frequency, measured in units of the identity radius R_{id}. Gravity is the macroscopic expression of systems tending to reduce this offset. Conservation of relational information over isotropic surfaces yields the inverse-square law as a derived result, not an assumption.

The gravitational source is the relational phase-offset density $\rho_{\mathrm{rel}} = \kappa\,\omega_m^2\,\rho_m$, where $\kappa = R_V^2 R_m/m_p$ is the relational closure volume. This leads to $\nabla^2\phi = 4\pi\alpha\kappa\omega_m^2\rho_m$ with $G = \alpha\kappa\omega_m^2$.

A central structural result is that the framework defines a *natural gravitational scale* constructed entirely from m_p, c, $\hbar$, and the $n = 4$ closure, without any gravitational input:

$$\frac{G}{\alpha} \;=\; \kappa\,\omega_m^2 \;=\; \frac{4\sqrt{15}\,\hbar c}{m_p^2}$$

The dimensionless coupling α is the relational ratio of the visible mode within the closure:

$$\alpha \;=\; \alpha_G \cdot \frac{\omega_V^2}{\omega_N\,\omega_m} \;=\; \frac{G m_p^2}{\hbar c} \cdot \frac{1}{4\sqrt{15}}$$

where $\alpha_G = G m_p^2/\hbar c$ is the standard dimensionless gravitational coupling. G and α are not independently free: both are linked through this scale, which is fixed by the harmonic partition alone.

The framework connects proton structure, valence chemistry, cosmic expansion, and gravitational interaction through a single harmonic closure principle spanning approximately 46 orders of magnitude. Three independent outputs emerge from the same machinery: the inverse-square law, Newton's constant, and the natural gravitational scale. Falsifiable predictions include composition-dependent deviations from the weak equivalence principle and scale-dependent effective gravitational coupling.

Keywords: emergent gravity; relational physics; harmonic partition; binary closure; phase offset; inverse-square law; unification; weak equivalence principle

Originality and Novelty Statement

What is novel. This work identifies a *natural gravitational scale* $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$ constructed entirely from hadronic constants and the $n = 4$ binary closure, without gravitational input. The dimensionless coupling α is expressed as the relational ratio of the visible mode: $\alpha = \alpha_G \cdot \omega_V^2/(\omega_N \omega_m)$. Together these show that G and α are not independent free parameters but are linked through the harmonic partition.

Previous emergent-gravity approaches (Verlinde [4], Jacobson [3], Padmanabhan [5]) reproduce Newtonian gravity qualitatively from thermodynamic analogies but leave G as a proportionality constant absorbed into entropy definitions. This model expresses G in terms of microscopically meaningful quantities and identifies the structural origin of the gravitational hierarchy.

How it differs from existing literature. The key distinction from thermodynamic approaches is structural: entropic and holographic frameworks assign temperature and entropy to macroscopic screens, but have no microscopic mechanism that determines the numerical value of G. Verlinde derives an entropic force $F = T\,\Delta S/\Delta x$ that recovers Newton's second law near a holographic screen, but G remains absorbed into the Bekenstein–Hawking entropy formula as a prior. The present framework assigns a specific relational closure volume κ to each nucleon's closure state — the product of an identity cross-section R_V^2 and a confinement depth R_m — making G explicitly dependent on hadronic-scale physics and therefore in principle sensitive to composition. This cross-scale quantitative closure is absent from all comparison frameworks listed in Table 5.

Acknowledged limitations. The coupling α is calibrated, not derived (one free parameter). The framework is valid only in the weak-field, stationary regime. A relativistic tensor completion remains open. The isotropy of Postulate 4 is stated as consistent with the closure symmetry but not yet derived from the generative algebraic structure. The selection rule fixing binary depth n for each system is not proven as a theorem. Prior applications of the harmonic partition are internal consistency results from the same research program and are not claimed as independent validations.

1 Introduction

Gravity remains the only fundamental interaction whose coupling constant G is imposed empirically rather than derived from deeper principles. While General Relativity provides an exquisite classical description, its conceptual origin — why mass curves spacetime with precisely this strength — remains unexplained. The question this paper addresses is: *can the value of G, and the inverse-square form of the gravitational force, be expressed as consequences of pre-existing microscopic structure, rather than treated as primitive inputs?*

Within the author's research program, a harmonic partition of system frequencies has been applied (not independently validated) to three domains:

Scale	**System**	**Harmonic ratio**	**Result**
Hadronic	Proton charge radius	$\omega_V^2 : \omega_m^2 = 1{:}15$	$r_p \approx 0.8412$ fm
Atomic	Valence stability (s-p)	$a : b$ prime or 2^n	Harmonic weight $P = a + b$
Cosmological	Hubble expansion	$V = \omega R$	$H_0 = \omega_V$ when $R = R_H$

Table 2: Prior applications of the relational projection framework. Internal consistency results, not independent validations.

This paper extends the same principle to gravity, adding two structural results to the program: the inverse-square law and Newton's constant, both emerging from the harmonic partition without separate assumptions.

2 Relational Foundations

2.1 The Harmonic Partition Postulate

[POSTULATE] 1 (Universal Harmonic Partition). Every stable relational system admits an orthogonal decomposition of its total frequency ω_N:

$$\omega_N^2 = \omega_V^2 + \omega_m^2 \tag{1}$$

where ω_V is the projection frequency (visible, relational mode), ω_m is the confinement frequency (inertial, informational mode), and $\omega_N = mc^2/\hbar$ for massive systems. **[POSTULATE]** This is treated as a physical constraint selecting stable configurations through rational ratios $\omega_V : \omega_m$. A derivation from more primitive axioms is an open question.

2.2 Binary Closure Principle

[POSTULATE] 2 (Binary Information Closure). A relational system achieves maximal stability when its harmonic partition satisfies a binary closure of depth n:

$$\frac{\omega_V^2}{\omega_N^2} = \frac{1}{2^n}, \qquad n \in \mathbb{N} \tag{2}$$

Observed ratios: $n = 2 \to \omega_V^2 : \omega_m^2 = 1{:}3$ (atomic valence); $n = 4 \to \omega_V^2 : \omega_m^2 = 1{:}15$ (hadronic). **[POSTULATE]** The selection rule fixing n is an open problem. Structural motivation based on spin-$\frac{1}{2}$ and the real representation dimension of $\mathfrak{su}(2)$ exists in prior work, but a derivation theorem does not yet exist.

2.3 Identity Radius as Derived Quantity

[DERIVED] The visible radius is defined operationally as $R_V \equiv c/\omega_V$. For a nucleon in closure at depth $n = 4$:

$$R_{\text{id}}^{(\text{nuc})} = R_V = 4\bar{\lambda}_p = \frac{4\hbar}{m_p c} \approx 0.8412 \text{ fm} \tag{3}$$

[HYPOTHESIS] This numerical proximity to the CODATA 2018 proton charge radius (0.8414 ± 0.0019 fm, [1]) and to the muonic hydrogen determination (0.84087 ± 0.00039 fm, [2]) is treated as a falsifiable consistency check of the $n = 4$ assignment, not as a confirmed independent prediction. For a composite body of N nucleons: $R_{\text{id}} = N^{1/3} R_{\text{id}}^{(\text{nuc})}$.

3 Gravitational Model: Derivation from Harmonic Closure

3.1 Relational Distance as Phase Offset

[HYPOTHESIS] Within this framework, the distance between two systems is not a primitive geometric quantity but an expression of their *relational phase offset*: the accumulated difference in identity-mode frequency, measured in units of each system's identity radius R_{id}.

> *The identity radius R_{id} is the minimum difference in trajectory between two systems such that one is not the other. It is defined operationally as the path-length difference at which two systems first become distinguishable — not a geometric container, but a relational threshold.*

For two systems A and B of the same closure type (same n), separated by coordinate distance r, the relational phase offset is:

$$\Phi(r) = \frac{r}{R_{\text{id}}}, \qquad R_{\text{id}} \equiv \frac{c}{\omega_V} \tag{4}$$

$\Phi(r)$ is dimensionless: it counts how many identity-radii of difference separate the two systems. At $r = R_{\text{id}}$, the systems are *one identity apart* — the minimum distinguishable separation. At $r \gg R_{\text{id}}$, there is large unresolved relational information.

[POSTULATE] 3 (Relational Difference Elimination). Physical systems tend to reduce their mutual relational phase offset. The gravitational interaction is the macroscopic expression of this tendency: the large-scale dynamics of systems minimizing the informational distance between their projected states.

3.2 Conservation of Relational Information

[POSTULATE] 4 (Isotropic Closure). The harmonic closure condition $\omega_N^2 = \omega_V^2 + \omega_m^2$ does not select a preferred spatial direction. Consequently, the relational phase offset of a source system propagates isotropically outward from its identity surface.
Open question: A derivation of this isotropy from the generative algebraic structure of the framework [11] remains future work. It is stated here as a postulate consistent with the symmetry of the closure condition.

[DERIVED] From Postulates 3 and 4, the total relational information emitted by a source is conserved over concentric surfaces of distinguishability:

$$\Delta\Phi(r) \cdot 4\pi r^2 \;=\; \Delta\Phi(R_{\rm id}) \cdot 4\pi R_{\rm id}^2 \;=\; \text{const} \tag{5}$$

$$\boxed{\Delta\Phi(r) \;=\; \Delta\Phi(R_{\rm id}) \cdot \frac{R_{\rm id}^2}{r^2}} \tag{6}$$

This is the origin of the inverse-square law. It is not assumed (cf. assumption A4 in v2.1 of this manuscript) but *derived* from conservation of relational information over isotropic surfaces, a consequence of Postulates 3 and 4.

3.3 The Gravitational Potential as Accumulated Phase Offset

[HYPOTHESIS] The gravitational potential $\phi(r)$ is the energy cost of maintaining a relational phase offset at distance r — equivalently, the accumulated offset that the system tends to eliminate:

$$\phi(r) \;=\; -\,\alpha\,\omega_m^2 \int_r^\infty \Delta\Phi(r')\,dr' \;\cdot\; \frac{M}{m_p} \tag{7}$$

The prefactor ω_m^2 converts the dimensionless phase offset into an energy density (the confinement frequency sets the energy scale of the identity mode). The factor M/m_p counts the number of nucleonic sources. This expression holds for a point source M; the extension to a distributed density ρ_m is developed in §3.4.

[DERIVED] Substituting Eq. (6):

$$\int_r^\infty \frac{R_{\rm id}^2}{r'^2}\,dr' \;=\; \frac{R_{\rm id}^2}{r} \tag{8}$$

$$\phi(r) \;=\; -\,\alpha\,\omega_m^2\,\frac{R_{\rm id}^2}{m_p} \cdot \frac{M}{r} \tag{9}$$

[DERIVED] Dimensional check: $[\omega_m^2] \sim [\mathrm{T}]^{-2}$, $[R_{\rm id}^2] \sim [\mathrm{L}]^2$, $[m_p^{-1}] \sim [\mathrm{M}]^{-1}$: right-hand side $\sim [\mathrm{L}]^2[\mathrm{T}]^{-2}$. ✓

3.4 Relational Closure Volume and Source Density

Equation (9) holds for a point source. To extend to a distributed source density ρ_m, we identify the *relational closure volume* of each nucleon — the volume through which it exchanges relational phase offset.

Note on prior versions. In v2.1, κ was defined as a spherical closure volume $\frac{4\pi}{3}R_V^3/m_p \approx 1.49 \times 10^{-18}$ m^3 kg^{-1}. However, the relational exchange is not isotropic between modes: the identity projects a transverse section R_V^2 toward other systems, while the confinement mode determines a penetration depth R_m in the orthogonal direction. The natural effective volume is therefore anisotropic — a product of transverse area and longitudinal depth — rather than a sphere: $\kappa = R_V^2 R_m/m_p \approx 9.19 \times 10^{-20}$ m^3 kg^{-1}. This differs from the spherical definition by a factor $\frac{4\pi}{3}R_V/R_m = \frac{4\pi\sqrt{15}}{3} \approx 16.3$, absorbed into the recalibrated α.

[HYPOTHESIS] Each nucleon in harmonic closure occupies a relational closure volume

$$v_{id} \;=\; R_V^2 \cdot R_m \tag{10}$$

where $R_V = c/\omega_V$ is the *identity surface* (the cross-section visible to another system) and $R_m = c/\omega_m$ is the *confinement depth* (how deeply the phase offset penetrates into the system's identity mode).

This volume is not the geometric size of the nucleon. It is the effective volume through which relational phase offset is exchanged:

- R_V^2: the transverse section of identity — what the system projects outward toward other systems.
- R_m: the longitudinal depth of confinement — how far another system's phase offset can penetrate inward.

[DERIVED] The specific relational closure volume (per unit mass) is:

$$\kappa \;\equiv\; \frac{v_{id}}{m_p} \;=\; \frac{R_V^2 \cdot R_m}{m_p} \;\approx\; 9.19 \times 10^{-20}\ \mathrm{m^3\,kg^{-1}} \tag{11}$$

[DERIVED] The relational source density and Poisson equation:

$$\rho_{\mathrm{rel}} \;\equiv\; \kappa\,\omega_m^2\,\rho_m \tag{12}$$

$$\nabla^2\phi \;=\; 4\pi\alpha\,\rho_{\mathrm{rel}} \;=\; 4\pi\alpha\,\kappa\,\omega_m^2\,\rho_m \tag{13}$$

[DERIVED] Dimensional check: $[\kappa\,\omega_m^2\,\rho_m] = [\mathrm{L^3M^{-1}}][\mathrm{T^{-2}}][\mathrm{ML^{-3}}] = [\mathrm{T^{-2}}]$. ✓

3.5 Newton's Constant as a Derived Quantity

[DERIVED] For a point mass M, Eq. (13) gives $\phi(r) = -\alpha\kappa\omega_m^2 M/r$. Comparing with Newton's $\phi_N(r) = -GM/r$:

$$\boxed{G \;=\; \alpha\,\kappa\,\omega_m^2 \;=\; \alpha\,\frac{R_V^2\,R_m}{m_p}\,\omega_m^2} \tag{14}$$

3.6 Calibration and Parameter Accounting

[CALIBRATION] For nucleonic matter in ground-state closure ($n = 4$), using CODATA 2018 [1]:

$$R_V = \frac{c}{\omega_V} = 4\bar{\lambda}_p \;\approx\; 8.412 \times 10^{-16}\ \mathrm{m} \tag{15}$$

$$R_m = \frac{c}{\omega_m} = \frac{4}{\sqrt{15}}\,\bar{\lambda}_p \;\approx\; 2.172 \times 10^{-16}\ \mathrm{m} \tag{16}$$

$$\kappa = \frac{R_V^2\,R_m}{m_p} \;\approx\; 9.19 \times 10^{-20}\ \mathrm{m^3\,kg^{-1}} \tag{17}$$

$$\omega_m^2 = \tfrac{15}{16}\,\omega_p^2 \;\approx\; 1.905 \times 10^{48}\ \mathrm{s^{-2}} \tag{18}$$

[CALIBRATION] Setting $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ (CODATA 2018) fixes $\alpha \approx 3.81 \times 10^{-40}$.

3.7 The Natural Gravitational Scale

[DERIVED] The product $\kappa\,\omega_m^2$ is constructible entirely from m_p, c, $\hbar$, and the $n = 4$ closure, with no gravitational input. Using $\lambda_p^3\,\omega_p^2 = \hbar c/m_p$ (exact):

$$\kappa\,\omega_m^2 \;=\; \frac{R_V^2\,R_m}{m_p}\,\omega_m^2 \;=\; \frac{4\sqrt{15}\,\hbar c}{m_p^2} \;\approx\; 1.751\times 10^{29}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} \tag{19}$$

[DERIVED] This is the *natural gravitational scale* of the $n = 4$ closure. It relates G and α without fixing either independently:

$$\boxed{\frac{G}{\alpha} \;=\; \frac{4\sqrt{15}\,\hbar c}{m_p^2} \;=\; 2^{n/2}\sqrt{2^n-1}\,\frac{\hbar c}{m_p^2}} \tag{20}$$

where the factor $2^{n/2}\sqrt{2^n-1} = (\omega_N/\omega_V)\cdot(\omega_m/\omega_V)$ is the product of the two modal frequency ratios of the closure. **[DERIVED]** This follows directly from the binary closure definition: $\omega_V = \omega_N/2^{n/2}$ gives $\omega_N/\omega_V = 2^{n/2}$; and $\omega_m = \omega_N\sqrt{1-2^{-n}}$ gives $\omega_m/\omega_V = 2^{n/2}\sqrt{1-2^{-n}} = \sqrt{2^n-1}$. Their product is $2^{n/2}\sqrt{2^n-1}$. For $n = 4$: $4\sqrt{15}$.

[DERIVED] Introducing the standard dimensionless gravitational coupling $\alpha_G \equiv Gm_p^2/\hbar c$, the coupling α admits an exact closed-form expression:

$$\boxed{\alpha \;=\; \alpha_G\cdot\frac{\omega_V^2}{\omega_N\,\omega_m} \;=\; \frac{Gm_p^2}{\hbar c}\cdot\frac{1}{4\sqrt{15}}} \tag{21}$$

Interpretation. α is not an independent free parameter: it is the standard gravitational coupling α_G weighted by the relational ratio $\omega_V^2/(\omega_N\omega_m)$ — the structural proportion of the visible mode within the harmonic closure. The numerator ω_V^2 is the visible mode squared — what the system projects outward. The denominator $\omega_N\cdot\omega_m$ is total times confinement — what the system retains internally. Their ratio is a fixed property of the $n = 4$ closure, not a free parameter.

What the framework establishes and what remains open. Eq. (20) is an exact identity derived from the harmonic partition with $n = 4$: the scale G/α is fixed by hadronic physics alone. Eq. (21) expresses α in terms of α_G and the closure ratios — it does not derive α_G independently. **[CALIBRATION]** One empirical input remains: either G or α_G must be supplied from measurement. Deriving α_G from the framework would make the gravitational scale fully predictive; this is listed as the program's primary open question.

Property	**Description**
Natural scale	$G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2 \approx 1.751\times 10^{29}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$. Fixed by $n = 4$ closure. No gravitational input.
α structural form	$\alpha = \alpha_G\cdot\omega_V^2/(\omega_N\omega_m)$. Relational ratio of the visible mode.
Empirical input	**[CALIBRATION]** One of $\{G, \alpha_G\}$ from CODATA 2018. Fixes $\alpha \approx 3.81\times 10^{-40}$.
Open question	Deriving $\alpha_G = Gm_p^2/\hbar c$ from the closure would close the program.

Table 3: Parameter accounting (v2.3).

3.8 Summary: Three Outputs from One Postulate

[DERIVED] The harmonic partition $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure ($n = 4$) yields three independent structural outputs from the same machinery:

1. **The inverse-square law** — derived, not assumed:
   conservation of relational information over isotropic surfaces $\Rightarrow \Delta\Phi(r) \propto R_V^2/r^2 \Rightarrow |\mathbf{g}| \propto 1/r^2$.

2. **The natural gravitational scale** — purely from closure:
   $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$, fixed by m_p, c, $\hbar$, $n = 4$. No gravitational input required.

3. **The structure of α** — relational ratio of the closure:
   $\alpha = \alpha_G \cdot \omega_V^2/(\omega_N \omega_m)$, the structural proportion of the visible mode within the harmonic partition.

4 Unification Across Scales

[DERIVED] All four physical domains follow from $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure:

Scale	System	n	ω_V	ω_m	Emergent property
Hadronic	Proton	4	$\omega_p/4$	$\sqrt{15}/4\,\omega_p$	Charge radius ≈ 0.8412 fm
Atomic	s-p valence	1,2	N_s freq.	N_p freq.	Harmonic weight P
Cosmological	Hubble flow	0 (lim.)	H_0	$\to 0$	Expansion law $V = H_0 D$
Gravitational	Newtonian	4 (nuc.)	c/R_{id}	ω_m	$G = \alpha\kappa\omega_m^2$; $1/r^2$ derived

Table 4: Quadripartite unification via $\omega_N^2 = \omega_V^2 + \omega_m^2$. Spans ~ 46 orders of magnitude. Gravitational entry now lists both outputs.

5 Testable Predictions and Falsifiability

5.1 Composition Dependence — Weak Equivalence Principle

[HYPOTHESIS] If $\omega_{m,t}$ (or $\kappa_t = R_{V,t}^2 R_{m,t}/m_p$) varies slightly with composition — e.g., via binding energy per nucleon — the model predicts small WEP violations. The Eötvös parameter:

$$\eta_{A,B} \approx \frac{\Delta(\omega_m^2)}{\omega_m^2} + \frac{\Delta\kappa}{\kappa} = \frac{\Delta(\omega_m^2)}{\omega_m^2} + \frac{2\Delta R_V}{R_V} + \frac{\Delta R_m}{R_m} \tag{22}$$

The second form shows explicitly that WEP violations arise from composition dependence of *both* the confinement frequency and the two modal radii.

FALSIFICATION CONDITION: This model predicts η should correlate with mean binding energy per nucleon, not only with atomic mass. Current limits from MICROSCOPE [6] give $\eta \lesssim 10^{-15}$. The model is falsified if η shows no composition-binding-energy correlation at next-generation experiment precision (STE-QUEST, MICROSCOPE follow-on). A quantitative estimate requires an explicit microphysical map from nuclear composition to ω_m and R_m.

5.2 Scale-Dependent Gravity and Dark-Matter Phenomenology

[SPECULATIVE] In diffuse systems, if closure is partial, the effective $\kappa(r) = R_V^2(r)\, R_m(r)/m_p$ becomes scale-dependent, giving $G_{\text{eff}}(r) = \alpha\kappa(r)\omega_m^2$.

FALSIFICATION CONDITION: Rotation-curve anomalies should correlate with informational density $\omega_m^2\rho_m$, not only with total mass distribution. This prediction distinguishes the framework from both ΛCDM and MOND [9, 10]. Important caveat: the r-dependence of κ is not derived from the postulates. This is a program note, not a genuine prediction.

6 Discussion

6.1 Comparison with Other Emergent Gravity Frameworks

Approach	**Source of gravity**	**Status of G**	$1/r^2$
Newton / Einstein	Mass / stress-energy	Fundamental	Assumed
Verlinde [4]	Entropy gradient	Proportionality	Assumed
Jacobson [3]	Horizon thermo.	Proportionality	Assumed
Padmanabhan [5]	Horizon/thermo.	Emergent (qual.)	Assumed
This work	Phase-offset density	$G = \alpha\, R_V^2 R_m\, \omega_m^2/m_p$	**Derived**

Table 5: Comparison with emergent-gravity frameworks. Note: "quantitative" refers to internal program consistency, not independent experimental confirmation.

Verlinde derives an entropic force $F = T\,\Delta S/\Delta x$ recovering Newton's second law near a holographic screen, where T is the Unruh temperature and ΔS is the entropy change associated with displacing a test mass. The approach is elegant and physically motivated, but T and ΔS are assigned to macroscopic holographic screens without a microscopic derivation of why G takes its observed value: G remains absorbed into the Bekenstein–Hawking entropy formula as a prior. Jacobson's derivation [3] faces the same issue: the Einstein equation emerges from $\delta Q = T\,\delta S$ applied to local Rindler horizons, but G enters through the identification of entropy with horizon area, which is itself proportional to $1/G$.

In contrast, the present framework assigns a specific relational closure volume $\kappa = R_V^2\, R_m/m_p$ to each nucleon's closure state, making G explicitly dependent on hadronic-scale physics. Critically, *all* prior approaches assume the $1/r^2$ form of the gravitational force from the outset. This framework derives it from conservation of relational information — providing a second structural output from the same machinery.

6.2 Physical Interpretation of α

[DERIVED] The structural result of §3.7 gives α a precise meaning within the framework:

$$\alpha = \alpha_G \cdot \frac{\omega_V^2}{\omega_N\,\omega_m} = \frac{Gm_p^2}{\hbar c} \cdot \frac{1}{4\sqrt{15}} \approx 3.81 \times 10^{-40}$$

The smallness of α does not reflect an arbitrary hierarchy. It is the product of two suppressions: the standard gravitational coupling $\alpha_G \approx 5.9 \times 10^{-39}$ (itself small, reflecting

the weakness of gravity relative to hadronic scales) and the relational ratio $\omega_V^2/(\omega_N\omega_m) = 1/4\sqrt{15} \approx 0.065$ — the fraction of the total closure structure that projects into the visible, relational sector.

Within this interpretation, gravity is weak not because spacetime curvature is intrinsically small, but because the visible mode carries only $\omega_V^2/\omega_N^2 = 1/16$ of the total harmonic partition energy, and its coupling to the confinement mode is further attenuated by $\omega_V/\omega_m = 1/\sqrt{15}$.

[CALIBRATION] What remains open is whether $\alpha_G = Gm_p^2/\hbar c$ itself can be derived from the closure framework. If so, the gravitational scale would be fully predictive. This is the program's primary open question.

6.3 Relation to the Hierarchy Problem

[SPECULATIVE] The hierarchy $\alpha \sim 10^{-40}$ reframes the gauge hierarchy problem (why $G_F \gg G_N$) as a question about the relational ratio of confinement information into macroscopic relational structure. Whether this reframing is more than a relabeling remains to be shown.

6.4 Cosmological Consistency Check

[SPECULATIVE] Using $M_U \sim 10^{53}$ kg [7] and calibrated α, the closure depth estimate is $n \sim 10^{13} \approx 2^{45}$. This is an assumption-dependent plausibility estimate, not a falsifiable prediction. Presented for transparency only.

6.5 Limitations and Future Work

Limitations:

- Weak-field, stationary regime only.
- α is calibrated, not derived.
- Relativistic completion remains future work.
- Isotropy of Postulate 4 is stated, not derived from the algebra.
- Physical correspondence with QFT not established.
- n-selection rule not proven.

Future work:

- Derive α from a higher-order closure condition.
- Derive isotropy of Postulate 4 from the generative algebra of [11].
- Develop a strong-field tensor formulation.
- Quantitatively fit galaxy rotation curves using $\kappa(r)$.
- Establish the n-selection theorem.

- Derive finite-size corrections to the phase-offset integral for $r \lesssim R_{\mathrm{id}}$, yielding quantitative coefficients β, γ in the near-closure potential. This requires a relativistic completion and is a prerequisite for neutron-star falsification targets.
- Map nuclear binding energy to ω_m and R_m variation for a quantitative WEP prediction.

7 Conclusion

We have presented a derivation of Newtonian gravity from relational projection principles in which neither the inverse-square law nor Newton's constant are assumed. The key elements are:

1. **[DERIVED]** Relational phase offset $\Phi(r) = r/R_{\mathrm{id}}$ as the fundamental measure of separation (Postulate 3).
2. **[DERIVED]** Conservation of relational information over isotropic surfaces $\Rightarrow$ $\Delta\Phi(r) \propto R_{\mathrm{id}}^2/r^2 \Rightarrow$ inverse-square law (Postulates 3–4). Derived, not assumed.
3. **[DERIVED]** Gravitational potential as accumulated phase offset $\Rightarrow \phi(r) \propto -M/r$.
4. **[DERIVED]** Relational closure volume $\kappa = R_V^2 R_m/m_p$ from the two modal scales of the harmonic partition.
5. **[DERIVED]** Poisson equation $\nabla^2\phi = 4\pi\alpha\kappa\omega_m^2\rho_m$ with $G = \alpha\kappa\omega_m^2$.
6. **[DERIVED]** Natural gravitational scale: $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$, fixed entirely by m_p, c, $\hbar$, $n = 4$. No gravitational input required.
7. **[DERIVED]** Structural form of α: $\alpha = \alpha_G \cdot \omega_V^2/(\omega_N\omega_m)$ — the relational ratio of the visible mode within the $n = 4$ closure. G and α are linked through the natural scale, not independently free.
8. **[HYPOTHESIS]** Falsifiable avenues: composition-dependent WEP deviations correlating with binding energy per nucleon, and scale-dependent effective gravitational coupling distinguishable from MOND in rotation curves.
9. **[CALIBRATION]** One empirical input: $\alpha_G = Gm_p^2/\hbar c$ from CODATA 2018. Deriving α_G from the closure is the program's primary open question.

"Gravity, like the size of the proton, the stability of atoms, and the expansion of the cosmos, is not a fundamental interaction but a manifestation of relational geometry seeking harmonic closure — the macroscopic tendency of systems to reduce the phase offset that separates them from each other."

LLM Disclosure Statement

Models used: OpenAI ChatGPT, Anthropic Claude, Google Gemini. Sessions used for adversarial validation were conducted logged-out with blank context.

How they were employed:

- Dimensional-analysis consistency checking across all equations.
- Algebraic stress-testing: presenting each postulate chain and requesting identification of unjustified steps or circular reasoning.
- Literature synthesis: identifying prior work in emergent gravity and equivalence-principle experiments.
- **Adversarial Cross-Validation (AACV):** a separate session prompted with a neutral review task using the contest rubric criteria.
- Structural refinement in response to community critique (r/LLMPhysics): the derivation of $1/r^2$ from conservation of relational information and the redefinition of κ as $R_V^2 R_m/m_p$ were developed in response to reviewer feedback requesting two independent outputs from the same machinery.
- Language revision and LaTeX formatting.

Estimated contribution: $\sim 35\%$ drafting/structure; $\sim 65\%$ human-directed conceptual development, postulate formulation, and iterative refinement.

Human oversight: All mathematical claims were verified by the author independently. LLM outputs were treated as drafting aids and consistency probes, not epistemic authorities. Citation details were cross-checked against primary sources.

Community acknowledgment: Feedback from r/LLMPhysics contributed to structural improvements. The v2.2 revision addresses the critique that the machinery produced only one output (G) while assuming the force law; it now derives both.

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Notation and Glossary

Symbol	Value / Form	Meaning
ω_N	$m_p c^2/\hbar$	Total (Compton) frequency of the system.
ω_V	$\omega_N/2^{n/2}$	Visible mode: the relational, projective frequency.
ω_m	$\omega_N\sqrt{1-2^{-n}}$	Confinement mode: the inertial, identity frequency.
R_V	$c/\omega_V = 4\bar{\lambda}_p$	Identity radius (transverse cross-section). Minimum path-length difference for two systems to be distinguishable.
R_m	$c/\omega_m = \frac{4}{\sqrt{15}}\bar{\lambda}_p$	Confinement depth (longitudinal penetration): how far a phase offset reaches into the identity mode.
κ	$R_V^2 R_m/m_p$	Specific relational closure volume (per unit mass). $\approx 9.19 \times 10^{-20}$ $\mathrm{m}^3\,\mathrm{kg}^{-1}$.
α	$\alpha = \alpha_G \cdot \dfrac{\omega_V^2}{\omega_N\,\omega_m}$	Universal dimensionless coupling: relational ratio of the visible mode within the harmonic closure. $\approx 3.81 \times 10^{-40}$.
α_G	$Gm_p^2/\hbar c$	Standard dimensionless gravitational coupling. $\approx 5.91 \times 10^{-39}$.
$\Phi(r)$	r/R_{id}	Relational phase offset at separation r. Dimensionless; $\Phi = 1$ at minimum distinguishable separation.
n	$n = 4$	Binary closure depth (nucleonic ground state). Fixes ratios: $\omega_V/\omega_N = 1/2^{n/2}$, $\omega_m/\omega_N = \sqrt{1-2^{-n}}$.

Table 6: Notation glossary. All quantities for nucleonic matter in ground-state closure ($n = 4$), CODATA 2018.